

BIOMIMETIC PROSTHETICS REPORT

S.ID: CLI:15263

REQUIREMENTS

Name: Kumar

Medical Condition: Traumatic Injury

Prosthetics Type: Ankle Ligament Reconstruction (Leg)

BIO-ANALYSIS

Material Composition: Synthetic Fiber

Age of Application Years: 4

Usage Frequency Times/Day: 3

Impact Resistance: 8

Wear Resistance: 7

Corrosion Resistance: 8

Temperature Resistance: 9

Integrity: 8

Weight: [0.2, 0.2, 0.2 ,0.2]

Characteristics: [8, 7, 8, 9]

Integrity Level (Prediction): 6.17

ACTUBIO

Actuation Mechanism: Pneumatic Actuators

Biocompatibility Factors: Osseointegration (bone connection)

Typical Force Output N: N/A (external power source)

Specific Surface Area: nan

Control Response Time: 10 - 50 (fast, depends on valve design)

Energy Density: 0.05 - 0.2 (estimated, under development)

Weight Actubio: 0.1 - 1 (very low)